MDB Search & Retrieval Specification v1.0

Search-first rules, retrieval behavior, workspace routing, BOM display logic, and tooling assembly explosion standards

Project: Mobile Drill Brain (MDB)
Document Type: Architecture / Product Specification
Version: 1.0
Date: July 18, 2026

Controlling Idea

MDB should reduce the number of clicks, searches, windows, and mental context switches required to answer a customer question.

The user should not have to know where data lives. MDB should interpret the search, preserve the active context, and route the user to the correct workspace.

1. Document Control

Document Name	MDB Search & Retrieval Specification v1.0
Relationship to Constitution	Operationalizes MDB Architecture Save Point v1.0 by defining search behavior and retrieval rules.
Primary Use	Guide future MDB database design, UI design, metadata tagging, GP export mapping, and prototype build decisions.
Status	Baseline specification. Future changes should be versioned.
Core Principle	Search first. Context preserved. Authoritative systems respected.

2. Purpose

This document defines how MDB search should behave. It converts the search-first architecture into software rules: what the user can search, how MDB interprets that search, which context should be preserved, which workspace should open, what source systems are authoritative, and what guardrails prevent false confidence.

MDB is not intended to be a generic document search box. It is a work accelerator for parts, tooling, rig sales, calibration, and troubleshooting calls. Search results should return usable workspaces and next actions, not just records.

3. Scope

In Scope for v1.0	Out of Scope for v1.0
Universal search and contextual search rules	Full GP replacement
Customer-card order history search	Automated order submission without GP
Serial-specific BOM routing and display standards	Duplicating proprietary engineering drawings into MDB
Tooling builder and expandable tooling assembly logic	Predictive failure analytics
Fleet authority rules for Mobile Drill vs competitor equipment	Automated pricing synchronization design details
Confidence/source labeling and ambiguity handling	Multi-user permissions and live access controls

4. Search Architecture Overview

Search-first Architecture Rule

MDB primary navigation is search. Menus and trees support the work, but the user should be able to start with a customer name, serial number, part number, description, tooling size, workflow phrase, symptom, or plain-English request.
Every meaningful search should resolve into a workspace or an actionable narrowing step.

```
User input
  -> Intent detection
  -> Active context check
  -> Source authority check
  -> Workspace routing
  -> Results + next action
  -> GP / customer / engineering / MDB knowledge as needed
```

5. Search Layers

Layer	Where It Appears	Searches First	Expansion Behavior
Universal Search	Always visible / MDB home screen	All indexed MDB entities	Routes to best workspace based on intent.

Customer Profile Search	Inside customer card/profile	Customer order history, fleet, contacts, notes	Expands to global catalog without leaving the customer context.
Workspace Search	Inside Rig Portal, BOM View, Tooling Builder, Troubleshooting, etc.	Current workspace scope	Expands outward while preserving the current workflow and quote context.

6. Active Context Bar

MDB should maintain an active context bar so the user and search engine both know what the current call is about. This prevents MDB from interpreting later searches as isolated, generic searches.

```
Customer: XYZ Drilling
Fleet Type: Mobile Drill / Competitor / Mixed / Unknown
Active Rig: Serial number, if selected
Active System: Rotary Box / HSA / Moyno Pump / Split Spoon / Core Barrel / etc.
Active Workflow: Parts / Tooling / Rig Sales / Calibration / Troubleshooting / Quote Builder
Confidence Source: Serial BOM / purchase history / customer confirmed / global catalog / needs verification
```

7. Searchable Entities

Entity Type	Examples	Primary Workspace
Customer	XYZ Drilling, US Timer, phone/contact/company	Customer Profile
Mobile Drill Serial Number	2025009, serial-linked fleet unit	Rig Portal
Rig Model	B57, B61, B59, etc.	Rig Spec / Recommendation Workspace
Part Number	GP PN, vendor PN, assembly PN, hardware kit PN	Parts Workspace or relevant BOM/tooling tree
Assembly	Drill Head, Rotary Table, Spindle Attachments, Breakout	BOM View / Rig Portal
Tooling Item	HSA auger, AWJ rod, split spoon, core barrel, cutterhead	Tooling Workspace
Connection / Compatibility Type	AWJ, NWJ, 2-key, HSA size, thread type	Tooling Builder
Component Model	Moyno 3L6, 3L8, pump family	Parts Workspace / Component Family
Workflow / KC	HSA setup, calibration workflow, sampler setup	Knowledge / Guided Workflow
Troubleshooting Symptom	Rotary will not turn, breakout slipping, pump issue	Troubleshooting Workspace
Production Schedule Entry	Next open build slot, unspoken-for rig	Rig Sales Workspace

8. Core Search Routing Table

Search Input	MDB Interpretation	Open / Show First	Required Context	Authority / Source
Customer name	Customer lookup	Customer Profile	None	Customer record / GP customer data
Serial number	Mobile Drill fleet unit	Rig Portal with BOM View option	Known serial	Serial-specific BOM / GP
Rig model	Model/spec or sales question	Rig Workspace / Recommendation	Customer goal if sales call	MDB rig knowledge + production schedule
Part number	Exact item lookup	Part Detail / Assembly Context	Serial if compatibility matters	GP item / MDB index
Assembly phrase	BOM/support area	Ask or use serial; open	Serial if Mobile Drill rig-	Serial BOM + MDB tags

		BOM section	specific	
Tooling size/configuration	Tooling item or builder path	Tooling Builder	Quantity, length, connection, application	MDB tooling catalog + GP inventory
Customer says “same as last time”	Repeat purchase lookup	Customer purchase history	Customer identity	Order history
Component model	Standalone component family	Component Parts Workspace	Model verification	Customer history + catalog + GP
Symptom phrase	Troubleshooting intent	Troubleshooting Workspace	Customer/fleet/serial as applicable	MDB knowledge + serial BOM where available
Calibration phrase	Calibration workflow	Calibration Workspace	Hammer/customer details	MDP workflow; GP only for invoicing

9. Context-Preserving Search Expansion

Context-Preserving Search Expansion

A search made inside a workspace should first search that workspace context. If no useful result is found, MDB should automatically expand outward without forcing the user to leave the current customer, rig, quote, or workflow.
The user should not have to back out of a customer profile to search a different auger size, a global tooling item, or a component part not previously purchased by that customer.

Current Context	User Searches	MDB Behavior
Inside XYZ customer profile	4.25 2-key	Search XYZ history first. If not found, show global 4.25 2-key auger options while keeping quote tied to XYZ.
Inside Rig Portal	bearing	Search serial-specific BOM first. Then allow model/global part search if needed.
Inside 3L6 Moyno context	rotor, stator, packing	Interpret as 3L6 Moyno rotor/stator/packing, not generic global terms.
Inside Tooling Builder	split spoon	Search current quote/customer history and tooling catalog while preserving the active customer and quote.

10. Active Context Locking

As a call develops, MDB should become more specific. Once the customer, rig, component model, tooling family, or assembly is confirmed, subsequent searches inherit that active context until the user clears or changes it.

```
Start: Customer says “Moyno pump parts”
Narrow: Customer says “3L6 or 3L8”
Confirm: Customer history or customer verification indicates 3L6
Lock Context: Customer + Moyno + 3L6 + Parts Lookup
Later Search: rotor / stator / packing
MDB Interprets: rotor/stator/packing for confirmed 3L6 Moyno pump
```

11. Confidence and Source Labels

MDB must show why it believes a result is correct. Confidence labels are not decoration; they are guardrails that prevent MDB from treating weak evidence as serial-verified fact.

Label	Meaning	User Action
Serial BOM Verified	Part/configuration found in Mobile Drill serial-specific BOM.	Strongest starting point; still verify field modifications when relevant.
Customer History Match	Customer previously purchased the item or related family.	Use as likely match; verify if equipment is competitor or uncertain.

Customer Confirmed	Customer verbally confirmed model/configuration/nameplate.	Useful for competitor fleet and non-rig-specific components.
Global Catalog Match	Found in MDB/GP tooling or parts catalog without customer/serial proof.	Quote only after compatibility is verified.
Needs Verification	MDB can narrow options but cannot prove the correct part.	Ask follow-up, request serial/model/photo/tag, or escalate.

12. Fleet Authority Rules

Search authority changes depending on whether the customer has a Mobile Drill fleet unit, competitor equipment, mixed fleet, or unknown equipment.

Fleet Situation	MDB Can Use	MDB Must Not Assume	Primary Verification Path
Mobile Drill Fleet	Customer fleet profile, serial number, serial-specific BOM, original configuration	That field changes never occurred	Serial BOM plus customer confirmation if wear/component may have changed
Competitor Fleet	Customer history, customer-provided model, photos/nameplate, global catalog	That a Mobile Drill BOM exists	Purchase history plus customer verification
Mixed Fleet	Both paths, depending on the specific unit involved	That the current item belongs to the Mobile Drill serial unless selected	Clarify equipment before quoting
Unknown Fleet	General search and narrowing questions	That search results are correct without confirmation	Ask who they are, what equipment, model, serial/tag, and application

Competitor Fleet Guardrail

For competitor equipment, MDB may use purchase history and customer-confirmed model numbers to suggest likely parts, but it must not present those suggestions as serial-verified unless a Mobile Drill serial-specific BOM exists for the selected rig.

13. Parts Search Rules

13.1 Serial-Dependent Parts Lookup

When a customer describes a rig component problem, and the required part depends on rig configuration, MDB should route to the customer/fleet/serial path before treating any part result as correct.

```
Customer says: “I am having issues with my rotary box on my B57.”
User asks: Who are you with? What is the serial number?
Search: serial number
MDB opens: Customer Card -> Fleet -> Selected Serial -> Rig Portal -> BOM View
User navigates: Drill Head / Rotary Table / Rotary Box assembly
Customer clarifies: top and bottom bearings on spindle side
MDB shows: matching assembly items, part numbers, quantities, related
races/cups/cones/seals/hardware, inventory/pricing links
```

13.2 BOM Display Standard

MDB should preserve the Mobile Drill BOM/manual hierarchy. It should not flatten serial-specific BOMs into a generic parts dump unless the user directly searches by exact part number.

```
Rig Serial Number
  -> Section
    -> Parent Assembly
      -> Child Assembly / Hardware Kit
        -> Item Number / Callout
          -> Part Number
```

```
          -> Description
          -> Quantity / UoM
          -> GP inventory/pricing where available
```

The user-provided Section J Drill Head and Section B Spindle Attachments samples confirm the parent-child pattern: serial-specific section, major assembly, child assembly or hardware kit, item/callout numbers, part numbers, descriptions, and quantities. MDB should model and display that structure rather than replacing it with a flat table.

13.3 Physical / Callout View

BOM View should support the way a customer describes physical location. When drawings or callouts exist, MDB should tie item numbers to the part list so the user can move from “top bearing,” “bottom bearing,” “spindle side,” “cover,” “seal,” “hardware kit,” or similar language to the likely items.

> **BOM Orientation Rule**
>
> Where Mobile Drill BOM/manual sections already follow physical top-to-bottom or assembly orientation, MDB should preserve that order.
> Physical orientation is a retrieval aid. It does not replace serial-specific part verification.

13.4 Non-Rig-Specific Component Parts

Some parts calls are not automatically rig-specific, such as a customer needing Moyno pump parts. MDB should not force a serial lookup if the customer knows and verifies the component model. If the customer is unsure and the component came on a Mobile Drill rig, the fleet/serial/BOM path becomes the best way to identify the original component.

```
Resolution order:
1. Customer-provided component model
2. Customer purchase history
3. Customer fleet profile
4. Mobile Drill serial-specific BOM/configuration, if applicable
5. Global component/parts database
6. Manual verification or escalation
```

14. Tooling Search Rules

14.1 Customer-History First When Customer Says “Same as Last Time”

When a customer says they need whatever they bought last time, their usual item, or a repeat purchase, MDB should search customer order history first before showing the global tooling catalog.

```
Customer: “I need some augers - whatever I bought last time.”
Search: Customer name
MDB opens: Customer Profile -> Purchase History -> Recent Auger Purchases
User verifies: “You most recently bought 2.25 2-key. Is that what you need?”
Customer corrects: “No, I need 4.25 2-key.”
Search inside profile: 4.25 2-key
MDB expands globally without leaving customer profile and opens Tooling Builder tied to that
customer.
```

14.2 Tooling Builder

Tooling searches should be builder-driven when the item depends on size, connection, length, footage, compatibility, or related components. MDB should help convert customer language into an item list.

```
Example:
Selected tooling: 4.25 2-key auger, 5 ft sections
Customer says: “about 50 ft”
MDB suggests: 10 x 5 ft sections
```

```
MDB prompts: cutterheads? rods? samplers? drive adapters? related components?
Confirmed, suggested, declined, and needs-verification items remain visually separate.
```

14.3 Confirmed vs Suggested vs Declined Items

Bucket	Meaning	Example	Rule
Confirmed	Customer explicitly wants it	10 x 5 ft 4.25 2-key augers	Can move to quote/order list
Suggested	MDB/user thinks customer may need it	cutterhead, rods, drive adapter, split spoons	Do not add until customer confirms
Declined	Customer was asked and said no	No cutterheads needed	Keep visible to avoid re-asking during the call
Needs Verification	Likely but not proven	customer's usual sampler when multiple options exist	Verify before quoting

15. Tooling Assembly Explosion Standard

Some tooling items are effectively assemblies. MDB should support vague search, progressive expansion, and reverse lookup from full description or exact part number. Expansion gives context and replacement breakdown; it must not hide orderable part numbers.

Part Number Visibility Rule

MDB shall display part numbers at every level of an expandable tooling assembly where a sellable or orderable item exists. Parent assemblies, complete tooling items, subassemblies, replacement components, wear parts, and hardware kits should all show their associated part numbers without requiring the user to open a separate record.

15.1 HSA Cutterhead Search

The cutterhead rule in this specification applies specifically to HSA cutterheads unless the user/search context indicates another family. MDB must not invent generic cutterhead categories. The actual styles, sizes, and compatibility paths must come from Mobile Drill tooling data, GP descriptions, confirmed BOM relationships, and verified compatibility rules.

```
Search: HSA cutterheads / cutterheads while inside HSA tooling context
MDB shows: available HSA cutterhead styles/families from real Mobile Drill data
User clicks: 4.25 2-key bullet style - PN shown immediately
Expand:
  -> bit holder - PN shown
  -> bits - PN shown
  -> hardware / wear parts - PNs shown where orderable
  -> compatibility notes and inventory/pricing links
```

15.2 Split Spoon Assembly Search

```
Search: split spoons
MDB shows: 18 in / 24 in
User clicks: 24 in
MDB shows: rod/thread compatibility options such as AWJ, NWJ, etc. based on actual catalog data
User clicks: AWJ
MDB shows: standard / tabbed if available
User clicks: tabbed
MDB shows: blunt shoe / sharp shoe / bag of catchers as compatible options
User clicks: sharp shoe
MDB shows: exact complete split spoon PN, full description, related replacement parts, and customer
history where applicable
```

The same item must also be reachable by full description or exact part number. If the user searches the exact PN, MDB should open the item and show the assembly path above and below it.

15.3 Core Barrel Assembly Search

```
Search: NQ core barrel
MDB shows: NQ core barrel complete - PN shown
User expands:
  -> outer tube - PN shown
  -> inner tube - PN shown
  -> head assembly - PN shown where applicable
  -> core lifter - PN shown
  -> core lifter case - PN shown
  -> reaming shell / bit / adapters / service parts as applicable
```

16. Rig Sales Search Rules

Rig sales search needs both direct mode and guided mode. If the customer knows the rig, MDB should open the rig/spec workspace directly. If the customer describes drilling method, depth, mobility needs, or end goal, MDB should open the rig recommendation workspace and narrow options by process of elimination.

Mode	Customer Language	MDB Behavior
Direct	“I am interested in a B57.”	Open B57 specs, options, known attachments, production schedule reference, lead-time discussion.
Guided	“I need a rig for 100 ft geotech/SPT work.”	Ask current rig, drilling methods, depth, mobility, desired timeline; narrow recommended rigs.
Pipeline Check	“When can I get one?”	Reference shared production schedule; show next unspoken-for rig or earliest open build start date.

17. Calibration Search Rules

Calibration remains primarily in the existing calibration workflow/MDP. MDB should support search and workflow guidance, but GP generally enters the picture at invoicing rather than during the technical calibration process.

```
Search: calibration / hammer calibration / customer hammer
MDB opens: Calibration Workspace
MDB supports: customer, hammer details, workflow status, report/reference, next actions
GP role: invoicing / formal transaction as required
```

18. Troubleshooting Search Rules

Troubleshooting searches should gather context and route to the right information without pretending to diagnose beyond available evidence. If the issue is tied to a Mobile Drill serial, MDB should use the serial-specific BOM and known configuration. If the issue exceeds user support scope, MDB should support escalation to the proper engineering department.

Customer Says	MDB Intent	Ask / Need	Show / Route
“Rotary will not turn.”	Symptom / rotary system troubleshooting	Customer, fleet, serial, hydraulic/mechanical symptoms	Troubleshooting workspace + rotary assembly/BOM + escalation path
“Breakout is slipping.”	Breakout system support	Serial if Mobile Drill, symptom detail, recent repairs	Breakout assembly, parts view, troubleshooting prompts
“Pump is not working.”	Component/system troubleshooting	Pump model, rig if applicable, customer history	Component family or serial BOM path

19. Metadata and Tagging Requirements

Every MDB record should be tagged for how a salesperson, customer, technician, or support person would actually search for it. Tags should include exact terms, common terms, abbreviations, model numbers, connection types, symptoms, assemblies, and related workflows.

Content Type	Required Metadata / Tags	Purpose
Customer	Name, aliases, contact names, phone/email, location, GP account, fleet units, order patterns	Find the customer quickly and preserve call context.
Serial/Rig	Serial, model, customer, section list, assemblies, original configuration, relevant manuals/BOMs	Open serial-backed Rig Portal and BOM View.
Part	PN, description, vendor PN, supersessions, assembly, compatible serial/model/family, related parts	Support exact and descriptive part lookup.
Tooling	Family, size, connection, length, style, complete PN, component PNs, compatibility, related items	Enable tooling builder and assembly explosion.
Knowledge Chapter	Plain terms, abbreviations, workflows, equipment family, symptoms, tooling links	Make KCs retrievable from live calls.
Troubleshooting	Symptoms, affected systems, required questions, escalation rules, related assemblies	Guide support without overclaiming.

20. Unknown and Ambiguous Search Handling

MDB should not guess silently. When a search could mean multiple things, MDB should show the likely paths and ask the smallest useful clarifying question.

Ambiguous Search	Possible Meanings	MDB Should Do
cutterheads	HSA cutterheads, other cutting tooling, customer-specific past purchase	Use active context. If no context, ask if HSA cutterheads are intended and show likely categories from actual data.
bearing	Many assemblies and models	Ask/use serial or active assembly before showing part results.
3L6 or 3L8	Moyno pump model uncertainty	Search both customer history and component family; prompt verification.
same as last time	Customer repeat purchase	Open customer order history grouped by likely category.
rotary box B57	Rig-specific parts/support issue	Ask for customer/serial and route to serial-specific BOM.
split spoon	Complete sampler or replacement parts	Open split spoon assembly tree with customer history and exact PN search support.

21. Source System Behavior

System / Source	Owns / Provides	MDB Behavior
GP	Customer records, serial BOMs, inventory, pricing, orders, invoice/order history	MDB references, indexes, and surfaces GP data; GP remains formal transaction engine.
Engineering System	Drawings, CAD, exploded views, proprietary engineering documentation	MDB references where possible; does not duplicate proprietary engineering files.
MDB Knowledge	Workflows, KCs, compatibility logic, recommendations, tags, assembly decision paths	MDB owns intelligence and retrieval layer.

Shared Production Schedule	Live rig build schedule	MDB may surface/open it; editing rights remain controlled outside MDB.
MDP / Calibration Workflow	Calibration process and technical workflow	MDB supports routing/search; GP used for invoicing as applicable.
Customer Input	Verbal confirmation, tag/nameplate, field modifications, current need	MDB records confirmation status and uses it as a confidence source.

22. Implementation Sequence

Efficient build order should prioritize the search skeleton before large-scale data intake. The goal is to define the rules once, then feed data into a stable structure.

Priority	Build Item	Reason
1	Search routing table and entity types	Defines what MDB recognizes and where searches go.
2	Customer Profile Search + order history search	High-value day-one workflow improvement.
3	Rig Portal + serial BOM tree prototype	Proves Mobile Drill fleet/BOM logic.
4	Tooling Builder MVP	Supports common auger/rod/sampler quote workflow.
5	Tooling Assembly Explosion MVP	Handles HSA cutterheads, split spoons, core barrels, and replacement parts.
6	Confidence labels and active context bar	Prevents wrong assumptions and preserves call flow.
7	Manual/BOM intake standard	Ensures future uploads map into the search architecture cleanly.

23. Architectural Decisions Log

Decision ID	Decision	Rationale
SR-001	MDB search returns workspaces, not just records.	The live call requires action, context, and next steps.
SR-002	Serial number is required when part identity depends on rig configuration.	Model-level lookup is insufficient for many Mobile Drill parts.
SR-003	Customer/profile search expands globally without leaving the profile.	Prevents backtracking and preserves quote/customer context.
SR-004	Active context locks as the call narrows.	Later searches should inherit confirmed customer, rig, system, model, and workflow.
SR-005	Competitor fleet results cannot be treated as serial-verified.	No Mobile Drill BOM authority exists for competitor equipment.
SR-006	BOM hierarchy must preserve Mobile Drill manual/GP assembly structure.	The existing parent-child and callout structure is useful and should not be flattened.
SR-007	Tooling assemblies use expandable explosion logic.	Complex tooling requires decision paths and replacement breakdowns, not flat lists.
SR-008	Part numbers display at every orderable level.	Expansion should reveal context, not hide sellable PNs.
SR-009	MDB must not invent tooling categories.	Tooling paths must come from verified Mobile Drill data and compatibility rules.
SR-010	Suggested related items are separate from confirmed order items.	Prevents accidental quoting or ordering of unconfirmed parts.

24. Future Parking Lot

These ideas remain valid but should not distract from v1.0 search/retrieval implementation.

- Live GP synchronization architecture and permissions
- Automated GP order creation from MDB quote builder
- Predictive maintenance and failure trends
- Inventory forecasting and reorder suggestions
- Customer buying pattern analytics beyond live-call retrieval
- Multi-user role-based editing and approval flows
- AI-assisted troubleshooting with engineering review controls
- ERP replacement path, if MDB eventually proves itself

25. Final v1.0 Guardrails

Non-Negotiables

1. Preserve context: never force unnecessary backtracking out of a customer, rig, quote, or workspace.
2. Respect authority: GP, engineering, serial BOMs, customer verification, and MDB knowledge each have defined roles.
3. Show confidence: always show whether a result is serial-verified, history-based, customer-confirmed, global, or needs verification.
4. Preserve hierarchy: BOMs and tooling assemblies should remain expandable and physically meaningful.
5. Do not invent compatibility: actual tooling paths and part relationships must come from verified Mobile Drill data.
6. Do not silently guess: when ambiguous, narrow with the smallest useful question.

26. Compatibility Enforcement Layer

Purpose

Search finds possible matches. Compatibility rules decide what is valid. This governing layer applies across all parts, tooling, BOM, customer-history, component, troubleshooting, calibration, and rig-sales retrieval behavior.

MDB shall never treat keyword similarity, customer history, or global catalog matches as enough to make an item quote-ready when compatibility, source authority, or model verification is still unresolved.

Compatibility Enforcement Rule

> MDB search may surface possible matches, but MDB shall only present an item as compatible, recommended, or quote-ready when it satisfies the relevant rig, tooling, connection, assembly, customer-context, and source-authority rules.

26.1 Governing Sequence

Every search result should pass through the same validation path before the user treats it as orderable or quote-ready:

```
User search -> Active context check -> Source authority check -> Compatibility check ->
Confidence/source label -> Quote-ready result OR needs verification / blocked / escalation
```

26.2 What Search Cannot Override

1. Serial-specific BOM authority when part identity depends on Mobile Drill rig configuration.

2. Established tooling compatibility rules for HSA size, key style, connection type, rod thread, sampler configuration, cutterhead style, bit/holder fit, core barrel assembly, and related replacement parts.

3. Fleet authority distinctions between Mobile Drill serial-backed equipment, competitor fleet equipment, mixed fleet, and unknown fleet situations.

4. Customer/model verification requirements for standalone component families such as Moyno 3L6 vs 3L8.

5. Complete assembly vs replacement-component relationships. Parent assemblies, child assemblies, wear parts, and hardware kits must remain linked to verified part numbers.

6. The rule that Mobile Drill tooling paths, styles, sizes, and part relationships must come from verified Mobile Drill data, GP item descriptions, confirmed BOM relationships, or established compatibility rules.

26.3 Validation Outcomes

Outcome	Meaning	Required User Behavior
Quote-ready	The result is compatible in the active context and supported by the proper authority source.	May move to quote/order list, subject to inventory, pricing, and customer confirmation.
Compatible - verify	The item appears compatible, but field changes, customer uncertainty, or weak source authority remain possible.	Verify serial, model tag, current equipment, or customer confirmation before quoting.
Possible match	Search found a plausible item, but compatibility context is incomplete.	Ask the smallest useful clarifying question before presenting as valid.
Blocked / not compatible	The item conflicts with known size, connection, serial, assembly, or family rules.	Do not quote unless compatibility is corrected or manually overridden by authorized review.
Escalate / needs review	MDB cannot verify the part, configuration, or troubleshooting path from available evidence.	Escalate to the proper department, request photo/tag/serial, or manually verify in GP/engineering resources.

26. Compatibility Enforcement Layer

26.4 Mandatory Compatibility Labels

MDB should display a compatibility/source label anywhere a result could be used for quoting, ordering, troubleshooting, or customer guidance.

Label	Use When
Serial BOM Verified	The item/configuration is found in the selected Mobile Drill serial-specific BOM.
Valid Tooling Compatibility Match	The item follows established tooling size, connection, thread, style, assembly, and replacement-part rules.
Customer Confirmed	The customer verbally verified the model, serial/tag, nameplate, current equipment, or needed configuration.
Customer History Supported	The item is supported by prior order history, but not necessarily current equipment verification.
Global Catalog Match Only	The item was found globally, but is not yet customer-, serial-, or model-verified.
Needs Verification	MDB can narrow the search, but cannot prove the correct part or compatibility from available context.

26.5 Examples of Enforcement

Search: 4.25 2-key - MDB may show only valid tooling families and parts that match the active tooling context and compatibility rules. Random descriptions containing 4.25 are not quote-ready.

HSA cutterhead -> 4.25 2-key bullet style - The complete cutterhead PN and expandable bit holder/bit PNs may be shown only when those relationships come from verified Mobile Drill data or confirmed compatibility tables.

Split spoon -> 24 in -> AWJ -> tabbed -> sharp shoe - MDB may display the exact complete split spoon PN only when each selected configuration step is compatible and supported by actual catalog data.

Competitor fleet Moyno 3L6 rotor/stator/packing - MDB may label the result customer-confirmed or customer-history-supported, but not serial-BOM-verified unless a Mobile Drill serial-specific BOM exists for the selected rig.

Bearing inside Rig Portal - MDB should search the selected serial-specific BOM and active assembly first. A global bearing match is not enough when the bearing depends on rig configuration.

26.6 Architectural Decisions Added

SR-011: MDB search is compatibility-governed. Search may discover possible results, but compatibility rules determine whether those results are valid, quote-ready, blocked, or need verification.

SR-012: Quote-ready results must carry an authority/confidence label. MDB must show whether a result is serial-verified, compatibility-verified, customer-confirmed, customer-history-supported, global-catalog-only, or needs verification.

Final Addendum Guardrail

> MDB is search-first, but not keyword-first. The purpose of search is to find the right path quickly; the purpose of compatibility enforcement is to keep that path correct.